

Course Syllabus: *Segnali Elettrofisiologici*

Master's Degree in Bioengineering | UNIMORE

- **Academic Year:** 2026/27
- **Course code:** BIOING_1 - 15806
- **Course info:** year 1, winter semester, 6 credits
- **Lecturer:** Prof. Eng. Michele GIUGLIANO, PhD
- **Format:** In-person lectures, chalk-and-talk, computational notebooks
- **Attendance:** Very strongly recommended

Syllabus: It is an outline of the topics taught in the course. It also contains additional information, designed to 1) preemptively answer students' questions, 2) list policy and mutual expectations and 3) the consequences of failing to meet them. It does not replace the official information about the course, published online at <https://unimore.coursecatalogue.cineca.it>, but extends it.

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1. Learning Objectives Students will be exposed to a selection of elementary topics in the genesis of electrical signals in nervous tissue, the biochemical and molecular signals of excitable cells. Students will understand the rationale behind a mechanistic dissection of the functioning of the mammalian nervous system at the cellular, microcircuit, and population levels. Students will ultimately gain expertise in this area and be able to apply their knowledge to capture an in-depth quantitative component in cutting-edge scientific publications.

Students will begin to make *quantitative* judgments and evaluations in the fields of *Electrophysiology* and Cellular Biophysics, analysing how significant progress can be made in these areas and where the availability of solid theoretical tools has proven extremely fruitful. Students will become fluent in this progression and skill.

Students will be introduced to a quantitative style of neural systems biophysics as an opportunity to expand their learning capabilities beyond “*enumeration*” in biology and into a new “*synthesis*” and “*analysis*” approach.

2. Prerequisite knowledge **Curricular requirements declared:** By enrolling in this Master's Degree program in Bioengineering for Innovation in

Medicine, students have declared they fulfil the curricular requirements set by UNIMORE and published on its website. This consists in *first-level degrees in classes L-8 (Electronic and Information Engineering) or L-9 (Industrial Engineering)*.

Past engineering background is not strictly compulsory, as long as students (from other backgrounds) declared they possess a **solid foundation in basic scientific disciplines**, with at least *20 ECTS (CFU) in the Scientific-Disciplinary Sectors (SSD) of **Mathematics and Computer Science**: MAT/02, MAT/03, MAT/05, MAT/06, MAT/07, MAT/08, SEC-S/01, SECS-S/02, SECS-S/06; and 16 ECTS (CFU) in the SSD of **Physical and Chemical Sciences**: FIS/01, FIS/03, FIS/07, CHIM/03, CHIM/06, CHIM/07.*

Furthermore, students enrolled in this Master's Degree program declared they acquired a **cultural background** in the scientific and technical disciplines typical of engineering and basic biomedical sciences, including *39 ECTS (CFU) in the following SSD: ING-IND/06, ING-IND/08, ING-IND/09, ING-IND/10, ING-IND/11, ING-IND/12, ING-IND/13, ING-IND/14, ING-IND/15, ING-IND/21, ING-IND/22, ING-IND/23, ING-IND/31, ING-IND/32, ING-IND/34, ING-INF/01, ING-INF/02, ING-INF/03, ING-INF/04, ING-INF/05, ING-INF/06, ING-INF/07, ICAR/01, ICAR/08, SECS-P/08; and 6 ECTS (CFU) in the following SSD: BIO/09, BIO/10, BIO11, BIO12, BIO14, BIO16.*

Prerequisite for this course: The curricular requirements broadly correspond to the foundational knowledge listed below (i.e., A, B, and C). These are also the prerequisites for **this** course.

A. Mathematics & Statistics

- **Basic Math, Functions & Algebra:** Elementary math, Algebra, Linear and nonlinear algebraic equations, functions of one variable, graph of a function, linear functions, exponentials, natural logarithms, and sinusoids/cisoids.
 - **SSD Reference:** MAT/05 (Analisi Matematica)
- **Calculus & Vector Analysis:** Derivatives (including composite functions/chain rule), definite and indefinite integrals, limits, first-order Taylor series expansion, partial derivatives, and spatial gradients; complex numbers, vector calculus.
 - **SSD Reference:** MAT/05 (Analisi Matematica)
- **Differential Equations:** First-order ordinary differential equations (ODEs) with constant coefficients, solving via homogeneous and particular solutions (forcing terms), and boundary/initial conditions (the Cauchy problem). Conceptual understanding of partial differential equations (PDEs).
 - **SSD Reference:** MAT/05 (Analisi Matematica) / MAT/07 (Fisica Matematica)
- **Linear Systems Theory:** The Dirac delta function, convolution integrals, impulse responses, and the fundamental concepts of low-pass and high-pass filters, Fourier transform, Laplace transform.
 - **SSD Reference:** ING-INF/03 (Telecomunicazioni) / MAT/05

(Analisi Matematica)

- **Probability & Stochastic Processes:** Probability, random variables, density distributions, stochastic processes, random fluctuations/noise.
 - **SSD Reference:** SECS-S/01 (Statistica) / MAT/06 (Probabilità)
- **Numerical Methods:** Discretization of continuous space and time, numerical integration methods (e.g., Euler, Runge-Kutta), and the concept of numerical stability.
 - **SSD Reference:** MAT/08 (Analisi Numerica)

B. Physics & Biophysics

- **Classical Mechanics & Thermodynamics:** Newton's law, mass, work, energy, conservative force fields, friction (Stokes' law), and the concepts of thermodynamic equilibrium and diffusion.
 - **SSD Reference:** FIS/01 (Fisica Sperimentale)
- **Electrostatics & Electromagnetism:** Coulomb's force, vector vs. scalar fields, electrostatic potential, electric fields, charge quantization, permittivity of a vacuum, relative permittivity of a medium, and the superposition principle.
 - **SSD Reference:** FIS/01 (Fisica Sperimentale)
- **Circuit Theory & Electronics:** Ohm's law, electrical resistance/conductance, capacitance, properties of parallel-plate capacitors, RC circuits, time constants (τ), Kirchhoff's Current and Voltage Laws, and lumped vs. distributed parameter models.
 - **SSD Reference:** FIS/01 (Fisica Sperimentale) / ING-INF/01 (Elettronica)

C. Chemistry & Physical Chemistry

- **Solutions & Stoichiometry:** Solutes vs. solvents, moles, molarity, Avogadro's number, conversion between volumetric concentration, surface density, and absolute number of particles.
 - **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica)
- **Ionic Species & Electrochemistry:** Properties of Na^+ , K^+ , Cl^- , and Ca^{2+} , electrochemical gradients, pH, and the dissociation of salts in aqueous media.
 - **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica)
- **Chemical Kinetics:** Reaction rates, reversible vs. irreversible reactions, the Law of Mass Action, and chemical equilibrium.
 - **SSD Reference:** CHIM/03 (Chimica Generale e Inorganica) / FIS/03 (Fisica della Materia)

Explicit review of prerequisite material in class: Although the very prerequisites for the enrollment in this Master's Degree program already cover the prerequisite for **this** course, the Lecturer is well aware of the diverse academic backgrounds of the students cohort and of their heterogeneous strengths and weaknesses. For such reason **the Lecturer will anyway explicitly review all the prerequisite material, during classes**. Please note that basic math, arithmetics, algebra, and calculus are covered through an asynchronous blended learning module ("Chapter 0") available online (<https://youtube.com/playlist?list=PL4ziDQ1W->

M22rbHT8dfz4IDmAp3z4dBuq) for all students and through a series of online (asynchronous) learning resources, presented in the overheads on Day 1 class.

Students with lack of prerequisite knowledge: Given the interdisciplinary nature of Bioengineering, students may occasionally discover significant gaps in their foundation knowledge throughout the classes. If a student identifies such gaps, **they are expected to contact the Lecturer as early as possible in the semester**. The Lecturer is fully committed to supporting student's extra learning, but **assistance can only be provided if these challenges are communicated proactively**. If the Lecturer is not made aware of specific deficiencies in advance, it will be impossible to offer the necessary support or tailored guidance later in the course.

Day 1, diagnostic non-graded questionnaires: To help the Lecturer calibrate the subsequent review of prerequisites during the forthcoming lectures, an anonymous Diagnostic Questionnaire will be provided on paper, during the first class. It is ungraded and it samples all the above topics. If a student choose not to complete it in class (or if they join the course late and fail to report to the Lecturer their own weaknesses in the prerequisite material), the Lecturer will assume administratively that "100% perfect mastery of all foundation mathematics, physics, and chemistry is already and satisfactorily represented in the students".

3. Core Class Content 0. Mathematical Preliminaries Mathematical functions and their graphs Derivatives and Integrals The Dirac's Delta and the Convolution Integral Ordinary differential equations and their solution Numerical solutions and the Euler forward method

1. Introduction Definition of "biological signal" Scientific motivations and innovation in medicine and neurobiology Examples of electrophysiological signals of the nervous system

2. EEG, Extracellular, and Intracellular Electrophysiological Signals Mentions of EEG and its applications in neurology Origins of EEG and its counter intuitive aspects

3. Biophysical Preliminaries Definition of ionic mobility in solution Capacitive and resistive properties of biological membranes Electrical Circuit Equivalent Models Definition of ionic fluxes and introduction of drift and diffusion

4. Neuroelectronics Thermodynamic equilibrium and the Nernst potential Transport currents through a semipermeable membrane Ohmic approximation to membrane permeability Goldman's equation and non-ohmic approximation Goldman-Hodgkin-Katz equations Definition and inference of "resting" membrane potential

5. Introduction to Cell Excitability Charge-balance equation Thevenin electrical equivalent circuit of a permeable membrane Intuitive account for permeability changes through time Introduction to Action Potentials (APs)

6. Further biophysical preliminaries Mass-action law, chemical reactions, and equations Kinetic schemes Non-deterministic microscopic interpretations of

kinetic schemes

7. Excitability according to Hodgkin & Huxley In-depth description of the HH model for AP generation Markov kinetic schemes for voltage-gated ionic channels Non-deterministic correlates of cell excitability

8. Synaptic transmission: chemical and electrical synapses Kinetic schemes for ligand-gated synaptic receptors Classification of synaptic receptors Excitatory and inhibitory synapses Electrical synaptic transmission and connectivity Reduced/simplified model of chemical synaptic transmission

9. Synaptic plasticity Short-term, homosynaptic plasticity Short-term depression and short-term facilitation Kinetic schemes for dynamics of ready-releasable synaptic resources Tsodyks-Markram model and its reduced version Long-term, heterosynaptic plasticity Spike-timing dependent plasticity, triplets of APs Considerations related to connectivity motifs

10. Extracellular Electrophysiological Signals Intra- and extracellular potential and spatially-distributed properties Derivation of cable equation and Rall's intuition Definition of electrotonus, solution of the cable equation Significance for spatial attenuation and temporal filtering of synaptic potentials Back-propagating action-potentials Synaptic Democracy

11. Volume Conductor Theory and Extracellular Potentials Point current sources and line current sources Contributions to extracellular potentials Discussion on applications of Volume Conductor Theory to common electrophysiological signals: spikes and LFPs

12. Analysis of Electrophysiological Signals Signals low-pass and band-pass filtering Peak-detection and spike-sorting Features extraction from intracellular signals Elementary spike-train analysis Brief mentions to Information Theory, Entropy, and Mutual Information for spike-train analysis

4. Study Materials, Textbooks, Resources

- **Primary References:** Hahn et al., (2024) "Electric Brain Signals"; Johnston & Wu, 1995 "Foundations of Cellular Neurophysiology"
 - Students with significant gaps in their foundational knowledge, are recommended consulting these additional reference material: *Weiss TF (1996) "Cellular Biophysics" vol. 1, MIT Press.*; *Molecular Biology of the Cell* (Alberts et al.); *From Neuron to Brain* (Nicholls et al., 1996); *Principles of Neural Science* (Kandel et al.) and *Eye, Brain, and Vision* (Hubel).
 - **Readings:** Some (compulsory) reading material, such as a condensed "Bignami-style" summary document on basic neuron and synapse physiology are available on GitHub.
 - **Online Resources and Study Material:** The course has a a) GitHub repository and b) a dedicated website for distributing educational materials and supporting students:
- 1) The **Course Website** (<https://mgiugliano.github.io/ePhysSignals>) provides a structured overview of the course, including the learning objectives,

the updated class schedule, and a detailed list of all the topics covered. In the “Resources” section, links to video tutorials, scientific literature, additional readings, mathematical prerequisites are provided.

- 2) The **GitHub Repository** (<https://github.com/mgiugliano/ePhysSignals>) serves as a central archive where lecture slides, Python notebooks, and reading materials are stored. Students must refer to this material to support their study and dive deeper into the technical and practical aspects of the module. The repository also features the refined (LLM-curated) transcripts of last year(s)’ classes. **Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.
- 3) **Anki Flashcards:** Assuming students might be Anki’s users (i.e., a spaced repetition flashcard program), in the GitHub repository of the course there are over 1,700 flashcards covering the course chapters to support their study. **Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.
- 4) A **Quiz Web App** (<https://mgiugliano.github.io/ePhysSignals/quiz>) is also offered to the students, featuring a very large database of typical exam/quiz questions (available only in Italian). Please keep in mind that these are generated in an unsupervised manner using Large Language Models (LLMs) and with no guarantees of correctness or completeness of this content.

“LLM Debugging” Challenge: As STEM scholars, students’ duty is the rigorous challenging of information received, not its passive acceptance. The LLM-curated transcripts, the LLM-generated Anki flashcards and quizzes are unsupervised and *may* occasionally contain *hallucinations*, contradictions or errors. The Lecturer calls all the students to treat these materials as raw, unreviewed literature. When you suspect a LLM hallucination, you are invited to formally “debug” it. To do so, you must submit a concise, written refutation to the Lecturer via email, that cites the context, isolates the specific error, and corrects it using first-principles reasoning and explicit, page-level citations from the book references (e.g., *Halnes et al.* or *Johnston & Wu*).

5. Grading Procedures

- **Overall grading:** The grade for *this* module will be weighted by its credits (i.e., 6 CFUs) and combined with the score the student achieves in all other modules, to determine the **final course grade**, rounded to the *closest integer*. Students must pass each module (i.e., *18/30* minimum). *30, Cum laude* in the overall scoring is awarded only if, after the CFU-weighted average and rounding, the score is *32/30*.
- Grades achieved by a student in any given module *expires* after *24 months*, requiring the students to pass that exam again.

- **Grading for this module:** The assessment of student performance in *this module* is designed to evaluate both continuous engagement with the material and ultimate mastery of the entire curriculum, structured around three distinct components:
 - 1) a primary **final written examination** (2.5h, no books, no notes, questionnaire closed-answers, exercises);
 - 2) a series of **brief continuous in-class assessments**, and
 - 3) an **optional** oral examination.

This approach ensures that students who cannot regularly attend lectures still have a viable pathway to passing the course with a high grade, while those who participate consistently and demonstrate exceptional comprehensive knowledge are duly recognized and rewarded.

To incentivize steady academic progress throughout the term, students are offered continuous in-class assessments. These take the form of brief digital quizzes, administered in-class at the very beginning of (most of the) lectures. These quizzers are accessible exclusively in the classroom via a projected QR code. These digital assessments evaluate the retention of topics covered in immediately preceding lectures. These brief assessments contribute cumulatively to a **maximum of 3 points over 30**. These points are strictly additive to the total score achieved on the final written examination. Missed quizzes cannot be recovered, compensated for, or made up under any circumstances.

The written examination requires registration on the esse3.unimore.it website for each session (i.e., January, February, June, July, September). This test evaluates the entirety of the course syllabus and serves as the foundational component of the student's final grade. Performance on this written examination alone can yield a *maximum score of 27 out of 30*.

Students are not required to have attended classes or completed the continuous assessments to take this exam, fulfilling the formal requirement that attendance is not strictly compulsory. The written exam is rigorously designed to assess deep theoretical understanding and practical application of the course concepts, ensuring a high standard of academic rigor for all enrolled individuals regardless of their physical attendance.

For example, a student who achieves the maximum score of 27 on the written exam and secures all 3 continuous assessment points will reach a full 30 out of 30.

An optional 20 min oral examination is available, in the form of an interview in front of a black board, upon request of each students, on case-by-case appointment. The appointment for the oral exam can be taken for dates in the exam session following the session when the written test has been taken by the student and anyway following the publication of the final written exam results. This oral assessment is the exclusive mechanism through which a student may achieve the highest possible distinction of **30 e lode (cum laude)**. Students electing to undergo the oral examination must be explicitly aware that it constitutes a comprehensive evaluation of their entire subject mastery. The oral examination is strictly bidirectional in its impact on the final course grade; a student's score may be increased, potentially leading to "cum laude", but it may

also be decreased if the oral performance demonstrates a lack of proficiency, reveals fundamental misunderstandings, or contradicts the mastery previously demonstrated in the written components. This structure ensures that the oral examination remains an endeavor reserved for those profoundly confident in their comprehensive command of the discipline.

Active LLM Debugging (Bonus): To reward the intellectual rigor and time required by the “LLM Debugging” Challenge, students who successfully submit a formal, textbook-backed refutation of a factual biophysical error found within the AI-generated course materials will be awarded up to **+1 bonus point**, which is strictly additive to their final exam grade.

- **Important Dates:** Exams sessions date (in January, February, June, July, September) are published on esse3.unimore.it. *Note: Students must register at least 10 days in advance, with sessions primarily in January/February).*
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6. Logistics and Policies

- **Class start and Breaks:** Lectures begin promptly **on the hour**. The *academic quarter* is not observed. Lectures breaks for **10 minutes every 50 minutes**.
- **Overheads:** Access to the updated overheads, used in class on a given day, is made available on GitHub (<https://github.com/mgiugliano/ePhy sSignals>) not less than 30 minutes before each class. Students may then focus on oral and chalk presentation by the Lecturer and on the interactions. Overheads are a *support* for the knowledge transfer not an end by itself. Contact hours in class serves to extend, explain, and integrated the information that may be referred to in the slides.
- **Videotaped lectures:** As the Master’s Degree program in Bioengineering is **not** an “online degree”, live lectures are neither systematically recorded nor streamed, unless noted otherwise. No provision for hybrid or online attendance is offered.
- Supporting working students, unable to attend the classes, or those dealing with severe and prolonged illness, the archival video recordings of past year(s)’ lectures are made available on the Lecturer YouTube channel (https://www.youtube.com/@michele_giugliano). Those older recordings are provided as a courtesy, with no implied guarantee of any correspondence in content, structure, or schedule with the current academic year’s classes. Thus, relying on those videotaped lectures as a substitute for in-person attendance is non-ideal, it is discouraged, and done entirely at the student’s own responsibility.
- **Questions:** Students are encouraged to actively ask questions and vigorously participate intellectually **during the class**. As they are early in their training career, students are definitely allowed to ask *stupid* questions. Overly shy students may approach the Lecturer before the start of a class or during the breaks for quick questions, being aware they are

preventing their peers to benefit from the “question and answer”. The Lecturer is passionate about passing his knowledge and fully committed to answer all questions at any time.

- **Attendance Policy:** Students are very strongly recommended to attend each class in person, to be on time and to avoid leaving before the end. Attendance tracking is done for 1) statistical purposes and, importantly, 2) for the in-class assessment and grading (see the Grading Procedures). Note: the Lecturer does **not** accept submission the in-class questionnaires by those students who are not physically in class.
- Attendance is tracked by the Lecturer for each class via the UNIMORE mobile app (<https://www.unimore.it/it/servizi/unimore-app>). Students **must** install this app on their mobile devices and use it to record their presence when they are in class. The Lecturer provides a unique code at the start of each class (e.g., 9PSTK) to be logged on the UNIMORE mobile app. Students exceptionally arriving late and thus *missing* the unique code communication must immediately alert the Lecturer.
- **Device Policy and Classroom Engagement:** Laptops, tablets, and smartphones are strongly discouraged during class to minimize distractions and promote active engagement. The Lecturer does not normally request that students leave their bags and devices at the class entrance, unless for the written exam. Devices inherently bring hundreds of other distractions that are always just one tap or one notification away.
- Students are requested to seriously commit to active participation in each class, demonstrating an investment of attention and focus that matches the Lecturer’s own investment of energy, effort, and care in delivering the lecture.
- **On Taking Notes:** Taking notes is strongly encouraged as a mean to identify and *digest* important concepts. There is consensus that hand writing notes on paper has a significantly better impact for long-term comprehension than typing on a laptop keyboard. Even gifted students will unlikely remember all the content of each class, without a permanent written record of the discussions and the additional explanation and insights provided by the Lecturer. Students should be also attentive to the questions of their colleagues: those are opportunities to learn and to clarify or check one’s understanding. Everyone should speak up and voice their doubts as they arise. Daily and weekly personal reviews, with serious study of the class material and discussions, is recommended. Indeed, studying Bioengineering at the University is a full-time job, an intense effort commitment, and an investment in one’s future.
- **Academic Honesty** in the advancement of knowledge requires that all students and Lecturers respect each other’s work integrity and recognise the importance of acknowledging and safeguarding intellectual property.
- As members of an academic community, students and faculty assume certain responsibilities.
- One of these responsibilities is to engage in honest communication.

- Academic dishonesty and cheating are serious violations of the trust upon which an academic community depends.
 - **Visit to the Lab:** Interested students are offered, upon appointment, a short tour of the Lecturer's labs.
 - The Lecturer can also be contacted at any times for information on stages, internships, and master thesis projects.
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7. Teaching Style, Language, and Philosophy

- **Format:** The teaching activity consists of frontal lectures where theoretical aspects and application examples are introduced and discussed. Furthermore, exercises and numerical computer simulations are presented in-class.
- **Language:** The Lecturer **prefers to deliver this course entirely in English and he offers it as an option**. This offer becomes a compulsory choice in the presence of *even a single international student*, to ensure full, open access to global knowledge.
- Lecturing in English transforms the classroom into a realistic professional microcosm, offering students a protected, low-risk environment to naturally absorb (and practice) technical terminology before entering the competitive high-tech industry or international research sectors. The Lecturer's proposal to use spoken English during classes is an offer of a strategic investment, designed to bridge the gap between *local* education and the *global* job market.
- When no international students are enrolled in the course, students **vote** on the language of the course during the first class: Italian or English. Students' choice carries definitive implications. The Erasmus+ bilateral partnerships established for the Master's Degree program by the Lecturer himself are founded on **strict reciprocity**; if UNIMORE does not routinely offer courses and traineeship in English, the mutual interest that sustains these agreements collapses. If this happens, the Lecturer withdraws his availability for his own personal sponsorship and any access to his own high-level international academic network for students' Erasmus+ placements. Those opportunities are instead exclusively reserved for students belonging to classes who chose the English track. This ensures UNIMORE (and the Lecturer) honours its exchange agreements and maintains the conditions for an inflow of international students, enriching UNIMORE. Students interested in an Erasmus+ placement and belonging to classes who opted for the Italian delivery of the course, need to secure their contacts and sponsorship through the personal networks of the other faculty members at UNIMORE.
- **Language used by students:** Regardless of the chosen lecture language, students can ask questions or seek clarifications speaking either English or Italian during class or office hours. They similarly retain full freedom to choose either Italian or English for their (optional) oral exam.

8. Contact and Communications

- **Contact:** Students can reach the Lecturer through email (michele.giugliano@unimore.it) or instant messaging (Teams) via

<https://teams.microsoft.com/l/chat/0/0?users=michele.giugliano@unimore.it>

- **Office hours:** The Lecturer provides explanations on the topics covered in class **only in person, and upon appointment** and not via Teams or email.
- Students should not hesitate to contact the Lecturer for explanations on any aspect of the course, or to request additional study material and pointers to the literature.
- Students are invited to use Teams to post their questions publicly, so that the entire class may try to answer them, with all benefiting from each interaction.
- **Teams App:** The only official channel used by the Lecturer to communicate to the students is the dedicated **Microsoft Teams group**. Teams is where announcements, posts, information, links, papers, and last-minute or urgent alerts are communicated to the students for their own benefit, interest, as well as for the administrative, logistic, and pedagogical aspects of the course.
- Students enrolled in this course should have been automatically added to the Teams group members by UNIMORE IT services. Students without a UNIMORE email account must **urgently** request the Lecturer, in written (email), to be added to Teams (i.e., *manually*).
- Students should install the Teams app on their device(s) and enable notifications.
- Teams is **not** meant as a unidirectional or passive communication channel. It serves as a **forum** where, **as a community**, students interact with the Lecturer and among themselves by asking/answering each others questions and exchanging information.
- **Student Responsibilities:** Any last-minute communication, sudden schedule changes, or emergency announcements regarding classes and exams will be posted by the Lecturer **exclusively** on Teams. It is entirely students' responsibility to stay informed.
- Missing an urgent update due to unchecked notifications will **not** be accommodated.
- Throughout their **entire** academic tenure as students at UNIMORE, individuals **must** check their Teams app and their unimore.it email account for new posts and messages **every working day, at least once per day**.